

Ben Harry Abraham

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RESEARCH INTERESTS

Quantum networking and quantum communication; entanglement distribution under realistic hardware and communication constraints; large-scale quantum network simulation; security of networked systems.

EDUCATION

Kean University <i>Master of Science in Computer Science</i>	Union, NJ Jan 2026 – May 2027 (expected)
Kean University <i>Bachelor of Science in Information Technology (Minor: Cyber Security)</i>	Union, NJ Sep 2022 – May 2025

RESEARCH EXPERIENCE

Graduate Research Assistant, TEEN Lab <i>Department of Psychology, Kean University</i> <i>Supervisor: Dr. Yu Sun Chung</i>	Sep 2026 – Present Union, NJ
- Serve as project manager for a developmental neuroscience study on how VR social stressors affect adolescent physiological and behavioral responses, coordinating multimodal data collection (EEG, actigraphy, salivary hormones, behavioral data) on a fixed timeline. - Support data cleaning, integration, and analysis in Python toward publication; completed human subjects and IRB training.	
Graduate Research Assistant <i>College of Business and Public Management, Kean University</i>	Dec 2025 – May 2026 Union, NJ
Designed and executed data cleaning and analysis pipelines in Python for Survey of Income and Program Participation (SIPP) datasets, spanning raw data acquisition and preprocessing across the 1996 to 2008 panels.	
Quantum Networking Researcher (REU) <i>Florida International University</i> <i>Supervisor: Dr. Deng Pan</i> <i>Funding: DoD, NSF</i>	May 2025 – Aug 2025 Miami, FL
- Designed and implemented quantum network simulations scaling to 1,200 nodes to evaluate how imperfect Bell State Measurements (BSMs) and classical communication delays affect end to end entanglement fidelity and throughput in quantum networking nodes. - Presented findings at the program's final research presentation.	
Socio-Economic Research Fellow <i>Kean University</i> <i>Funding: NSF</i>	Jan 2024 – May 2025 Union, NJ
Developed and implemented machine learning models in Python (logistic regression and additional statistical techniques) to predict contingent worker classification as part of an NSF funded research team.	
CAHSI Cybersecurity Researcher (REU) <i>CAHSI REU Program, Kean University</i> <i>Funding: NSF</i>	Feb 2024 – Nov 2024 Union, NJ
Conducted mentored cybersecurity research on malware prevention; authored a research plan, tracked progress, and presented findings as a research poster at the GMIS conference.	
IoT Research Assistant <i>Florida International University</i> <i>Supervisor: Dr. Xyuy Wang</i> <i>Funding: DoD, NSF</i>	May 2023 – Aug 2023 Miami, FL
Identified and evaluated cost-effective hardware and toolkits for a deep learning based radio fingerprinting system for IoT device authentication, enabling the collection of a reliable device-fingerprint dataset used to train the ML models.	

PRESENTATIONS

- Ben Abraham Mitigation strategy against ransomware attacks affecting health care systems. Research poster, GMIS Conference, Leander, TX, November 2024.
- Ben Abraham. Quantum Networking: Evaluating Entanglement Swapping Strategies by Simulation.” Final research presentation, Quantum Networking REU, Florida International University, Miami, FL, August 2025.

INNOVATION PROGRAMS

Co-Entrepreneurial Lead, Team SAGEM <i>NSF I-Corps Hub (Northeast Region)</i>	Jan 2025 – Feb 2025
Led customer discovery and market validation for a cognitive health product; conducted and analyzed 30+ customer interviews and delivered cohort presentations.	

PROFESSIONAL EXPERIENCE

Real Time Crime Center Data Analyst

Sep 2024 – Jan 2025

New Jersey State Police

Trenton, NJ

- Monitored and analyzed 9+ real time intelligence feeds, license plate recognition systems, and surveillance databases; produced intelligence reports, data visualizations, and crime pattern analyses for local, state, and federal partners.

End User Field Services Technician

Jan 2024 – Jun 2024

NJ Transit

Maplewood, NJ

- Provided on site hardware, software, and network support; configured and deployed workstations and supported user training.

Network Intern

Aug 2022 – Sep 2022

Cisco

New York, NY

- Deployed 16 Meraki MR76 and MR86 Wi-Fi 6 access points with Cisco engineers across VIP and outdoor zones for the Global Citizen Festival in NYC.

Founder

Jan 2019 – Present

Dynamic.IO (dynamicio.net)

- Founded and operate a software studio delivering web and mobile applications for clients, overseeing development, deployment, and client collaboration end to end.
- Built free community cybersecurity tools, including `metadataanalyzerv2.vercel.app`, `passwordgenanalyser.vercel.app`, and `genhash.vercel.app`.

TECHNICAL SKILLS

Research: Quantum network simulation, entanglement fidelity analysis, radio frequency fingerprinting, IoT security

Languages: Python, C++, JavaScript, TypeScript, SQL

ML & Data: PyTorch, TensorFlow, NumPy, pandas, statistical modeling, data cleaning and pipeline design

Security & Networking: Penetration testing, Cisco networking, Wi-Fi 6, RFID

HONORS AND COMPETITIONS

- Kean University Hackathon, 2nd place (2022): built an RFID door lock on an Arduino Uno (C++) with custom 3D printed parts
- NJIT National CTF, top 20% nationally (2023, 2025); GMIS National Conference CTF, top 25% nationally (2024)

SERVICE AND AFFILIATIONS

- Judge, SkillsUSA Tech Apps Competition, Thomas Edison High School, New York, NY
- Member, IEEE; Member, ACM (Kean University chapter)